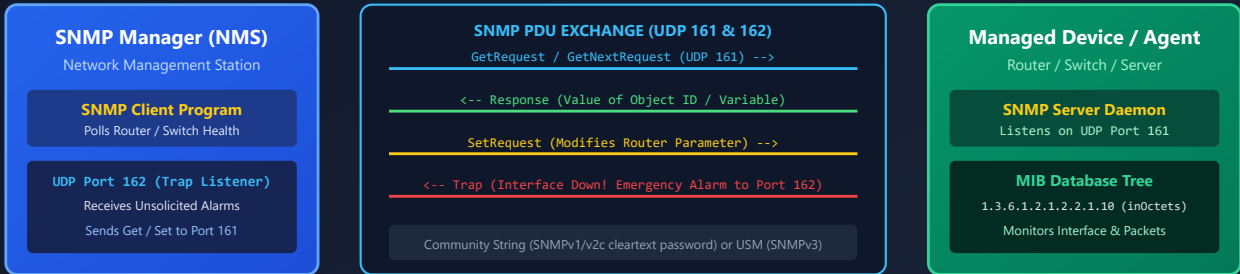


Module 09: Network Management and SNMP Architecture

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SIMPLE NETWORK MANAGEMENT PROTOCOL (SNMP) FRAMEWORK

Manager-Agent Paradigm, SMI Syntax, MIB Object Hierarchy, UDP Ports 161/162 & PDUs



THE NETWORK MANAGEMENT TRIAD: SMI vs MIB vs SNMP (SLIDE 65 ANALOGY)

SMI (Grammar / Syntax Rules)

Analogy: Syntax of C / Java language. • Defines naming conventions using ASN.1. • Defines basic data types (INTEGER, STRING) • Encodes packets via BER (Basic Encoding Rules). MIB (Variable Database) • **Analogy:** Declared variables in program. • Inverted tree structure of managed objects. • Every parameter has a unique Object ID (OID). • e.g., sysUpTime, ifInOctets, ipInReceives. SNMP (Communication Protocol) • **Analogy:** Function calls & message passing. • Defines packet structure & PDU headers. • **3 Core Tasks:** Querying variables (Get), Modifying state (Set), Asynchronous alarms (Trap).

1. SNMP Operational Architecture

Simple Network Management Protocol defines a framework for managing heterogenous devices in an internetwork across a manager-agent architecture.

Transport Layer Port Allocation

UDP Port 161: Managed Agent port accepting polling queries (GetRequest, SetRequest).

UDP Port 162: Manager NMS station port listening for unsolicited asynchronous alarms (Traps).

2. Network Management Triad (SMI vs MIB vs SNMP)

Component	Scope & Semantics	Structural Output
SMI	Grammatical specification and typing syntax	ASN.1 data type definitions (INTEGER, OCTET STRING, OID)
MIB	Structured database of all monitored system variables	Inverted tree of OIDs (e.g., 1.3.6.1.2.1.1 system group)
SNMP	PDU formatting and message passing protocol	Packet frames exchanged over UDP sockets

3. The Three Primary Management Functions

- **Status Polling (GetRequest / GetNextRequest):** Periodic inspection of interface traffic and CPU load.
- **Remote Configuration (SetRequest):** Modifying routing parameters and port operational states.
- **Event Notification (Trap):** Unsolicited emergency alarm reporting critical hardware failures.